



Modelling Financial Markets

Seminar 20: Path Signatures and Realised Volatility — Hands-on



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ AUDUSD: Australian dollar / US dollar exchange rate (FX pair code)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIC: Bayesian Information Criterion
- ▣ CI: Confidence Interval
- ▣ CL: Crude oil (WTI light sweet) futures, NYMEX ticker
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CVX: Chevron Corporation (NYSE ticker)
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: E-mini S&P 500 futures (CME ticker)
- ▣ EURUSD: Euro / US dollar exchange rate (FX pair code)
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ GBPUSD: British pound / US dollar exchange rate (FX pair code)
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ OLS: Ordinary Least Squares
- ▣ OPEC: Organization of the Petroleum Exporting Countries
- ▣ PRDS: Positive Regression Dependence on a Subset (of p-values)
- ▣ QLIKE: Quasi-LIKElihood (loss function)

Acronyms used in this material (2/2)

- ▣ RV: **Realised Variance**
- ▣ SHAR: **Semivariance HAR model**
- ▣ SHW: **Sherwin-Williams Company (NYSE ticker)**
- ▣ USDCAD: **US dollar / Canadian dollar exchange rate (FX pair code)**
- ▣ USDJPY: **US dollar / Japanese yen exchange rate (FX pair code)**
- ▣ VOLARE: **VOLatility Archive for Realized Estimates (database)**

Today's Question and Required Outputs

- Question of the seminar: **does the shape of the recent volatility path help to forecast realised variance, and how do we test that honestly?**
- Route
 - ▶ A1, A7, A9 on paper: a signature step by step, the QLIKE loss, kernel weights
 - ▶ B1, B3 on JPM: the Lévy area as a predictor and the HAR family, with HAC errors
 - ▶ B5: Sig-LK against log-HAR with DM, GW and a bootstrap interval; B7: Holm and BH over 50 assets
 - ▶ C2: find the errors in an AI answer
- Required outputs (checklist on the last core slide): the A answers, one figure and one table for each of B1, B3, B5, B7, the corrected C2 code, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); background: Lecture 20 and Chapter 9
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1	how does a signature record the order of moves?	Solved	–
A7	which forecast minimises QLIKE, which errors cost more?	Solved	–
A9	how does the temperature trade similarity for sample size?	Solved	–
B1	does the Lévy area predict next month's RV of JPM?	Solved	–
B3	do the SHAR and HARQ effects hold for JPM?	Solved	–
B5	does Sig-LK beat log-HAR for JPM, one day ahead?	Solved	–
B7	how many wins survive Holm and BH over 50 assets?	Solved	–
C2	what is wrong in an AI answer?	Proposed	A1, A7, B5
Extension packs	A2–A6, A8, A10, B2, B4, B6, B8, C1 (optional, after the seminar)	both	named on each task

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: verification and inference on VOLARE data, each ending with an interpretation question; Part C: open questions and AI critique

Data Manifest

File	Symbols	Sample	Used in
realized_variance_stocks.csv	JPM (all 40 stocks in C1)	2 Jan 2015 – 30 Jan 2026	A5, B1, B3, B5, C1
realized_variance_futures.csv	ES (E-mini S&P 500), CL (crude oil)	28 Sep 2009 – 30 Jan 2026	B2, B4, B6
ch20_results.json	50 assets, results of Lecture 20	one-sided DM p -values	B7, B8

- Columns of the VOLARE files [Cipollini et al. \(2026\)](#), one row per asset and trading day
 - ▶ rv5: realised variance from five-minute returns, $RV_t = \sum_i r_{t,i}^2$; rsp5, rsn5: its positive and negative parts (semivariances)
 - ▶ rq5: realised quarticity, the scale of the measurement error of RV; open_price, close_price: for the open-to-close log return r_t
- VOLARE is free for research at volare.unime.it: download the three realised-variance files and set their folder in the first notebook cell

Setup: Run and Verify

```
# notebook section "Signatures, models and tests"
# defines load_asset(kind, symbol): columns
# rv, rsp, rsn, rq and r = ln(close/open)
a = load_asset('stocks', 'JPM')
print(len(a), a.index[0].date(), a.index[-1].date())
print(pd.DataFrame({'RV_x_1e4': 1e4 * a.rv,
                    'r(%)': 100 * a.r}).head(3))
```

- Expected output: 2786 trading days, 2015-01-02 to 2026-01-30

date	RV $\times 10^4$	r (%)
2015-01-02	1.463	-0.22
2015-01-05	1.855	-2.56
2015-01-06	2.974	-2.77

- RV of 1.46×10^{-4} is a daily volatility of $\sqrt{1.46 \times 10^{-4}} = 1.2\%$
- If your count or first value differs, check the folder, the file and the symbol

Notation Used in Every Exercise

- RV_t : five-minute realised variance of day t ; $r_t = \ln(\text{close}_t / \text{open}_t)$: open-to-close log return
- HAR averages: $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$, $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$; log-HAR regresses $\ln RV_{t+1:t+h}$ on $\ln RV_t$, $\ln RV_t^{(w)}$, $\ln RV_t^{(m)}$, with $RV_{t+1:t+h}$ the mean RV over the next h days
- Path X with increments Δ_k ; signature levels $S^{(1)} = \sum_k \Delta_k$ and $S^{(2)}$ (a $d \times d$ matrix); Lévy area $A^{12} = (S^{12} - S^{21})/2$
- Loss $\text{QLIKE}(RV, F) = RV/F - \ln(RV/F) - 1$ for a forecast F ; loss difference $d_t = \text{QLIKE}_t^{(1)} - \text{QLIKE}_t^{(2)}$; DM statistic $t = \bar{d} / \sqrt{\hat{\Omega}_{\text{NW}}/T}$ with a Newey–West (HAC) long-run variance
- Kernel weights $w_\tau = e^{-\gamma \delta_\tau} / \sum_s e^{-\gamma \delta_s}$ for signature distances δ_τ ; Kish effective sample size $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$
- One worked line: $RV = 1$, $F = 2$: $\text{QLIKE} = 0.5 - \ln 0.5 - 1 = 0.193$

A1 [Solved]: A Depth-2 Signature, Step by Step — Task

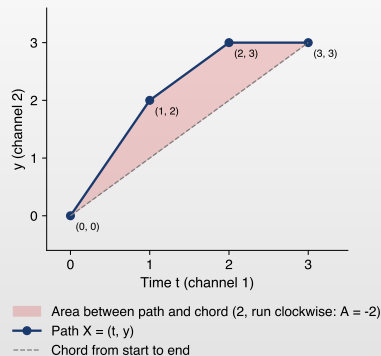
- **Question:** how does a depth-2 signature record *when* a series moved, not only how much?
- **Inputs**
 - ▶ A series takes the values $y = (0, 2, 3, 3)$ on days $t = 0, 1, 2, 3$; no data file
 - ▶ The time-augmented path $X_t = (t, y_t)$ is linear between consecutive days (channel 1 = time, channel 2 = y)
- **Tasks**
 - (a) Write the three increments $\Delta_1, \Delta_2, \Delta_3 \in \mathbb{R}^2$ and compute level 1, $S^{(1)} = \sum_k \Delta_k$.
 - (b) Use Chen's identity to show that $S^{(2)} = \sum_k \frac{1}{2} \Delta_k \Delta_k^\top + \sum_{j < k} \Delta_j \Delta_k^\top$, then compute this 2×2 matrix.
 - (c) Check the result with the shuffle identity $S^i S^j = S^{ij} + S^{ji}$ for $i = j = 1$ and for $i = 1, j = 2$.
 - (d) Compute the Lévy area $A^{12} = (S^{12} - S^{21})/2$ and draw the path with its chord.
- **Report:** $S^{(1)}$, $S^{(2)}$ and A^{12}
- **Interpretation:** what does the sign of A^{12} say about when the series rose?
- Notebook: section A1

A1 [Solved]: Solution (1/2) — Levels 1 and 2

- (a) Increments: $\Delta_1 = (1, 2)$, $\Delta_2 = (1, 1)$, $\Delta_3 = (1, 0)$; level 1: $S^{(1)} = (3, 3)$, the total change in time and in y
- (b) One segment has $\text{Sig} = (1, \Delta, \frac{1}{2}\Delta\Delta^\top, \dots)$; Chen multiplies the segments, and level 2 of the product collects every square term once and every ordered pair $j < k$ once
- Squares: $\frac{1}{2} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1.5 & 1.5 \\ 1.5 & 2.5 \end{bmatrix}$
- Ordered pairs: $\Delta_1\Delta_2^\top + \Delta_1\Delta_3^\top + \Delta_2\Delta_3^\top = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$
- Sum: $S^{(2)} = \begin{bmatrix} 4.5 & 2.5 \\ 6.5 & 4.5 \end{bmatrix}$

A1 [Solved]: Solution (2/2) — Shuffle Check and Lévy Area

- (c) Shuffle, $i = j = 1$: $S^1 S^1 = 9 = 2S^{11} = 2 \cdot 4.5$; $i = 1, j = 2$: $S^1 S^2 = 9 = S^{12} + S^{21} = 2.5 + 6.5$: both hold
- (d) Lévy area: $A^{12} = (2.5 - 6.5)/2 = -2$
- Geometric check (chart): the path lies above its chord $y = t$; the shaded polygon has area 2 and is run clockwise, hence the minus sign
- Interpretation: with channels (time, y), a negative area means that y rose *early* in the window; the same net change spread evenly would give $A^{12} = 0$

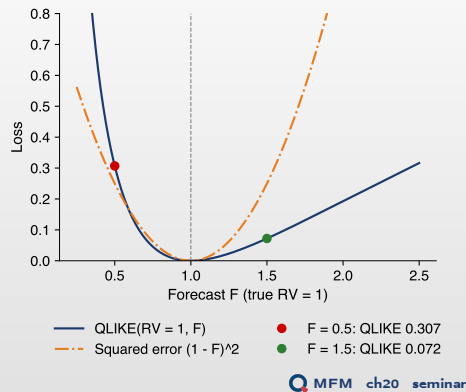


A7 [Solved]: QLIKE: Minimiser and Asymmetry — Task

- **Question:** which forecast minimises expected QLIKE, and which errors does QLIKE punish more?
- **Inputs:** $\text{QLIKE}(RV, F) = RV/F - \ln(RV/F) - 1$ with realised proxy $RV > 0$ and forecast $F > 0$; $\mathbb{E}_t =$ expectation given the information at the forecast origin
- **Tasks**
 - (a) Differentiate $\mathbb{E}_t[\text{QLIKE}(RV, F)]$ with respect to F and solve the first-order condition.
 - (b) Check the second-order condition.
 - (c) For $RV = 1$, compute QLIKE and the squared error for $F = 0.5$ and for $F = 1.5$, and plot both losses against F .
- **Report:** the minimiser, the four losses and the chart
- **Interpretation:** which error does QLIKE punish more, and why does this suit risk management?
- **Notebook:** section A7

A7 [Solved]: Solution

- (a) $\mathbb{E}_t[\text{QLIKE}] = \mathbb{E}_t[\text{RV}]/F + \ln F - \mathbb{E}_t[\ln \text{RV}] - 1$;
 $\partial/\partial F = -\mathbb{E}_t[\text{RV}]/F^2 + 1/F = 0 \Rightarrow F^* = \mathbb{E}_t[\text{RV}]$
- (b) $\partial^2/\partial F^2 = 2\mathbb{E}_t[\text{RV}]/F^3 - 1/F^2 = 1/F^{*2} > 0$ at F^* : a minimum; F^* depends only on $\mathbb{E}_t[\text{RV}]$, so with a conditionally unbiased proxy the *expected-loss* ranking is robust [Patton \(2011\)](#)
- (c) $\text{QLIKE}(1, 0.5) = 2 - \ln 2 - 1 = 0.307$;
 $\text{QLIKE}(1, 1.5) = 0.667 + \ln 1.5 - 1 = 0.072$; squared errors 0.25 and 0.25
- Interpretation: QLIKE punishes under-prediction of variance about four times more than the same over-prediction; under-predicted risk is the costly error (too little capital, breached VaR)

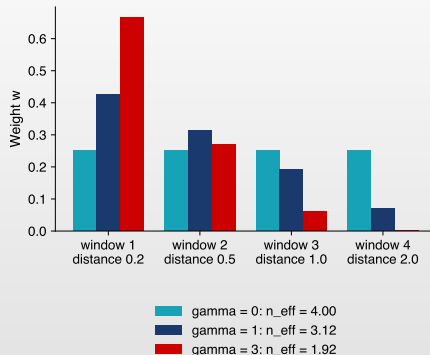


A9 [Solved]: Kernel Weights and Effective Sample Size — Task

- **Question:** how does the kernel temperature γ trade similarity against the number of observations?
- **Inputs**
 - ▶ Four past windows with signature distances $\delta = (0.2, 0.5, 1.0, 2.0)$ to the current window
 - ▶ Weights of Gu et al. (Eq. 12): $w_\tau = e^{-\gamma\delta_\tau} / \sum_s e^{-\gamma\delta_s}$; Kish effective sample size $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$
- **Tasks**
 - (a) Compute the four weights for $\gamma = 1$.
 - (b) Compute the four weights for $\gamma = 3$.
 - (c) Compute n_{eff} in both cases and compare with equal weights ($\gamma = 0$).
- **Report:** two weight vectors, two n_{eff} , a bar chart
- **Interpretation:** what is gained and what is lost when γ increases?
- **Notebook:** section A9

A9 [Solved]: Solution

- (a) $e^{-\delta} = (0.819, 0.607, 0.368, 0.135)$, sum 1.929:
 $w = (0.425, 0.315, 0.191, 0.070)$
- (b) $e^{-3\delta} = (0.549, 0.223, 0.050, 0.002)$, sum 0.824:
 $w = (0.666, 0.271, 0.060, 0.003)$
- (c) $n_{\text{eff}} = 4$ for $\gamma = 0$, 3.12 for $\gamma = 1$ and 1.92 for $\gamma = 3$
- Interpretation: a higher temperature focuses on the most similar window (less bias from dissimilar periods) but halves the information (more variance); in the lecture $n_{\text{eff}} \approx 816$ of 1000: little is lost, and little changes



B1 [Solved]: Does the Lévy Area Predict Volatility? (JPM) — Task

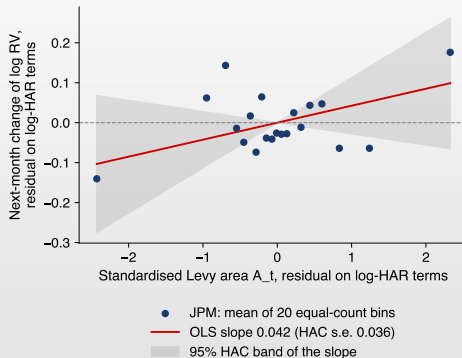
- **Question:** does the order of price and volatility moves over the last month predict JPM's volatility next month, beyond log-HAR?
- **Inputs**
 - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; columns `rv5`, `open_price`, `close_price`
 - ▶ Regressor: A_t = Lévy area of the 22-day path (cumulative r , $\ln RV$) ending at day t , divided by its standard deviation
 - ▶ Target: $y_t = \ln RV_{t+1:t+22} - \ln RV_t^{(m)}$, next month's log RV minus the past month's (consecutive targets overlap in 21 of 22 days)
- **Tasks**
 - (a) For every day, build the path of the last 22 days, compute its depth-2 signature and $A_t = (S^{12} - S^{21})/2$, then standardise A_t .
 - (b) Regress y_t by OLS on a constant, the three log-HAR terms and A_t , with Newey–West standard errors and 22 lags.
 - (c) Test H_0 : coefficient of $A_t = 0$ at 5%, two-sided, and compare the R^2 with and without A_t .
 - (d) Draw a binned scatter of y_t against A_t , both residualised on the log-HAR terms.
- **Report:** coefficient, HAC s.e., t , decision, both R^2 , the chart
- **Interpretation:** why 22 lags, and is the gain economically relevant?
- Notebook: section B1

B1 [Solved]: Solution (1/2) — Steps

- (a) Signature of each 22-point window with channels (return, $\ln RV$); A_t is negative in 46% of the windows (compatible with, but not identifying, leverage-type sequences)
- (b) 2743 usable days; coefficient of $A_t = +0.042$, HAC s.e. 0.036
- (c) $t = +0.042/0.036 = +1.18$; $|t| < 1.96$: do not reject H_0 at 5%; R^2 : 28.0% without A_t , 28.6% with A_t
- Why 22 lags: consecutive targets share 21 of their 22 days, so the errors are serially dependent even under H_0 ; persistent RV can extend the dependence, so a check with 44 lags is advisable

B1 [Solved]: Solution (2/2) — Chart and Interpretation

- (d) Chart: bins scatter around zero; the HAC band of the slope contains the horizontal line
- Relevance: half a point of R^2 next to the 28% explained by the log-HAR terms is negligible
- Interpretation: for JPM the order of moves between price and volatility adds no reliable information about next month's volatility



B3 [Solved]: HAR, SHAR and HARQ for JPM — Task

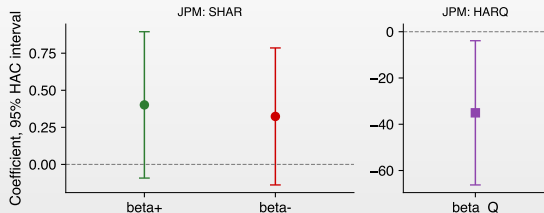
- **Question:** do bad volatility (SHAR) and the measurement error of RV (HARQ) change next-day forecasts for JPM?
- **Inputs**
 - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; columns rv5, semivariances rsp5, rsn5, quarticity rq5; target: next-day RV in levels, full sample
 - ▶ SHAR [Patton & Sheppard \(2015\)](#): $RV_{t+1} = \beta_0 + \beta^+ RV_t^+ + \beta^- RV_t^- + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}$
 - ▶ HARQ [Bollerslev, Patton & Quaedvlieg \(2016\)](#): HAR plus the term $\beta_Q RV_t (RQ_t^{1/2} - \overline{RQ}^{1/2})$; $V =$ Newey–West (5 lags) covariance matrix of the estimates
- **Tasks**
 - (a) Estimate HAR [Corsi \(2009\)](#) by OLS and report the three slopes with HAC standard errors.
 - (b) Estimate SHAR and test $H_0 : \beta^- = \beta^+$ against $\beta^- > \beta^+$ with $t = (\hat{\beta}^- - \hat{\beta}^+) / \sqrt{V_{--} + V_{++} - 2V_{+-}}$.
 - (c) Estimate HARQ and test $H_0 : \beta_Q = 0$ against $\beta_Q < 0$; plot β^+ , β^- and β_Q with 95% HAC intervals.
- **Report:** estimates, t , p , decisions, the coefficient chart
- **Interpretation:** do the two effects of the literature hold for this stock?
- Notebook: section B3

B3 [Solved]: Solution (1/2) — Estimates and Tests

- (a) HAR (2 764 days): $\beta_d = 0.367$ (0.080), $\beta_w = 0.494$ (0.149), $\beta_m = -0.026$ (0.095)
- (b) SHAR: $\beta^+ = 0.401$ (0.252), $\beta^- = 0.324$ (0.236); difference -0.078 , $t = -0.17$, one-sided $p = 0.566$: do not reject H_0
- (c) HARQ: $\hat{\beta}_Q = -35.062$ (15.904), $t = -2.20 < -1.645$: reject H_0 at 5% one-sided

B3 [Solved]: Solution (2/2) — Chart and Interpretation

- Chart: the two SHAR intervals are wide and overlap (the semivariances are highly correlated for one stock); the HARQ interval lies below zero
- Interpretation: on noisy days HARQ trusts the day's RV less, and this holds for JPM; the good/bad asymmetry is a panel result [Patton & Sheppard \(2015\)](#), not detectable for JPM alone



- SHAR beta+ (positive semivariance)
- SHAR beta- (negative semivariance)
- HARQ beta_Q

B5 [Solved]: Sig-LK vs log-HAR for JPM, One Day Ahead — Task

- **Question:** does the signature method of the lecture (Sig-LK) forecast JPM's next-day RV better than log-HAR?
- **Inputs**
 - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; columns `rv5`, `open_price`, `close_price`
 - ▶ Design of the lecture: rolling window of 1000 days, re-estimation every 5 days, path (time, $\ln RV$, cumulative r) over 22 days, depth 3, two-step LASSO with BIC, kernel temperature $c = 0.5$
- **Tasks**
 - (a) Produce one-day forecasts of Sig-LK and log-HAR from 27 Jan 2020 to 29 Jan 2026 (1511 days) and compute the mean QLIKE of each.
 - (b) Compute the ratio of the two mean losses and its 95% CI with the stationary bootstrap (mean block length 10, 1000 replications).
 - (c) Run the DM test (HAC variance, bandwidth $\max(h, \lceil T^{1/3} \rceil)$) and the conditional GW test with instruments $(1, d_{t-1})$ [Giacomini & White \(2006\)](#).
 - (d) Plot the cumulative loss difference $\sum_{s \leq t} d_s$.
- **Report:** losses, ratio and CI, t , p , decisions, the chart
- **Interpretation:** is there any evidence that the signature adds value for JPM?
- **Notebook:** section B5

B5 [Solved]: Solution (1/2) — Losses and Tests

- (a) Mean QLIKE: Sig-LK 0.1726, log-HAR 0.1727; on average 4.2 regressors selected, effective sample size 851 of 1000
- (b) Ratio 1.000; bootstrap 95% CI [0.956, 1.047] contains 1
- (c) DM: $d_t = \text{QLIKE}_t^{\text{Sig-LK}} - \text{QLIKE}_t^{\text{log-HAR}}$, $t = \bar{d} / \sqrt{\hat{\Omega}_{\text{NW}} / T} = -0.02$, one-sided $p = 0.493$: do not reject equal accuracy
- GW: Wald statistic $2.85 \sim \chi^2(2)$, $p = 0.241$: the instruments $(1, d_{t-1})$ give no significant evidence that the loss difference is predictable

B5 [Solved]: Solution (2/2) — Cumulative Loss Difference

- (d) Upward segments: log-HAR better; downward segments: Sig-LK better; the largest moves sit in March 2020 and April 2025
- A handful of stress days decides the sign of the total: report the cumulative path, not only the mean
- Interpretation: no evidence; the two forecasts are practically identical, and the CI rules out gains or losses larger than about 5%



— Cumulative QLIKE(Sig-LK) - QLIKE(log-HAR), JPM, $h = 1$

B7 [Solved]: Holm and BH, Step by Step — Task

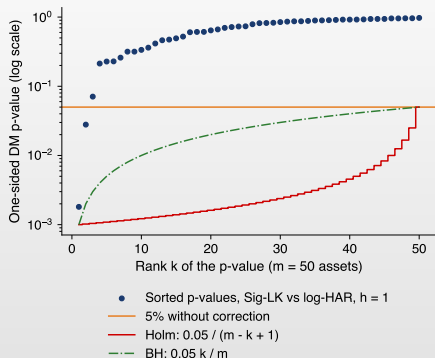
- **Question:** after correcting for 50 tests, on how many assets does Sig-LK beat log-HAR one day ahead?
- **Inputs**
 - ▶ Lecture 20 results (ch20_results.json): one-sided DM p -values for “Sig-LK better than log-HAR” at $h = 1$ (QLIKE), one per asset; family = the 50 assets
 - ▶ The five FX pairs, sorted: EURUSD 0.071, USDJPY 0.213, USDCAD 0.336, AUDUSD 0.718, GBPUSD 0.817
 - ▶ Holm (FWER 5%): compare the k -th smallest $p_{(k)}$ with $0.05/(m - k + 1)$ and stop at the first non-rejection; BH (FDR 5%) [Benjamini & Hochberg \(1995\)](#): reject the k smallest for the largest k with $p_{(k)} \leq 0.05 k/m$
- **Tasks**
 - (a) Apply Holm to the five FX p -values ($m = 5$).
 - (b) Apply BH to the same five.
 - (c) Repeat both procedures for all 50 assets, count the rejections and plot the sorted p -values against the three thresholds.
- **Report:** rejections raw, Holm, BH; the chart
- **Interpretation:** what may we claim, and about how many assets?
- Notebook: section B7

B7 [Solved]: Solution (1/2) — Steps

- (a) Holm, $m = 5$: $p_{(1)} = 0.071$ vs $0.05/5 = 0.01$: no rejection, stop; Holm-adjusted p : 0.354, 0.852, then 1
- (b) BH: thresholds 0.01, 0.02, 0.03, 0.04, 0.05; every $p_{(k)}$ is above its threshold: no rejection; the smallest BH-adjusted p is 0.354
- (c) 50 assets: 2 raw rejections; Holm: the smallest, ES with $p = 0.002$, is above $0.05/50 = 0.001$: stop, 0 rejections; BH: $p_{(1)} > 0.001$ and $p_{(2)} = 0.028 > 0.002$, no k qualifies: 0

B7 [Solved]: Solution (2/2) — Chart and Interpretation

- Chart: only the lowest points cross the 5% line; none reaches the Holm staircase or the BH line
- Holm is valid under any dependence; BH needs independent or positively dependent (PRDS) p -values [Benjamini & Yekutieli \(2001\)](#)
- Interpretation: no asset-level improvement survives either correction; ES is at most a lead for a new, pre-registered study



C2 [Proposed]: Critique an AI Answer — the Answer

□ **Prompt** sent to an AI assistant: “rv holds the daily 5-minute RV of one stock. Build the signature LASSO of Gu et al. for next-day log RV and evaluate it with QLIKE on the last 30% of the sample.”

□ **Answer:** the code below and the conclusion

```

lrv = np.log(rv.values)
S = np.array([sig_path(lrv[t-21:t+1, None], depth=3)      # 22-day window
              for t in range(21, len(lrv) - 1)])
y = lrv[22:]                                              # next-day log RV
lam = LassoCV(cv=5).fit(S, y).alpha_                     # choose lambda
cut = int(0.7 * len(y))
m = Lasso(alpha=lam).fit(S[:cut], y[:cut])
f = np.exp(m.predict(S[cut:]))
qlike = lambda a, b: a / b - np.log(a / b) - 1
print(qlike(f, np.exp(y[cut:])).mean())

```

□ **Conclusion of the AI:** “The signature LASSO has a lower QLIKE than HAR on the test set, so path signatures capture the shape of volatility and improve the forecast.”

C2 [Proposed]: Critique an AI Answer — Task

Solution discussed in class

- **Question:** which errors in the answer would change the conclusion?
- **Model:** A1 (time augmentation), A7 (QLIKE), B5 (evaluation without look-ahead); the code contains **at least three** errors
- **Tasks**
 - (a) Find at least three errors; for each, name the line and explain in one sentence why it is wrong.
 - (b) For each error, describe a check that reveals it without knowing the answer (a unit test, a timing test, a known case).
 - (c) Assess the exponential back-transform and whether the code is the two-step weighted method of Gu et al.
 - (d) Write the corrected code and compare it with log-HAR on the same split, with a DM test.
 - (e) Log each error as in [AI_ERRORS.md](#): request, answer, error, how found, correction.
- **Interpretation:** does the conclusion follow even if the code were correct?
- AI code usually runs without an error message: the mistakes are in the method, not in the syntax

Submission Checklist and Team Repository

Hand in before the next seminar

- A1, A7, A9: the matrices, losses and weights, with the three charts
- B1, B3, B5, B7: one figure and one table each, as in the solutions
- C2: the list of errors, the checks, the corrected code and its DM test against log-HAR
- One interpretation sentence per exercise

Team repository

- One Quantlet folder: B5 for a second stock of your choice, with the chart
- Log AI use in `AI_USE.md` and caught errors in `AI_ERRORS.md`
- Extension packs are optional; C1 is a possible project start

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Signature algebra	A2, A3 [Solved], A4	A1
2. HAR as an AR(22)	A5 [Solved], A6	B3, Wald tests
3. Losses and weights	A8, A10	A7, A9
4. Other assets	B2 (ES), B4 (CL), B6 (ES, $h = 5$)	B1, B3, B5
5. Multiple testing at one month	B8	B7
6. Project seed	C1, a level-aware kernel	B5, Lecture 20 (kernel weights)

- Each exercise has its own notebook section with starter code; within a pack, work in the order given
- Proposed exercises name their model: the solved exercise that uses the same technique

A2 [Proposed]: Signature and Shuffle Identity — Task

Solution discussed in class

- ▣ **Question:** what is the signature of a three-segment path in the plane, and which channel moved first?
- ▣ **Model:** A1
- ▣ **Inputs:** A path in $\mathbb{R}^2$ through $(0, 0) \rightarrow (2, 1) \rightarrow (3, -1) \rightarrow (4, 2)$, linear between points
- ▣ **Tasks**
 - (a) Write the increments and compute level 1.
 - (b) Compute level 2 with the formula of A1 (squares plus ordered pairs).
 - (c) Verify $S^1 S^2 = S^{12} + S^{21}$ and compute the Lévy area A^{12} .
- ▣ **Report:** $S^{(1)}$, $S^{(2)}$, A^{12} ; **Interpretation:** which channel moved first, on balance?
- ▣ **Notebook:** section A2

A3 [Solved]: Chen's Identity and Path Reversal — Task

- **Question:** what is the signature of a path run there and back, and why does this matter for time augmentation?
- **Inputs**
 - ▶ Path $X : (0, 0) \rightarrow (1, 0) \rightarrow (1, 2)$ from the lecture, with $S^{(1)} = (1, 2)$ and $S^{(2)} = \begin{bmatrix} 1/2 & 2 \\ 0 & 2 \end{bmatrix}$
 - ▶ $\overleftarrow{X}$ is the same path run backwards: $(1, 2) \rightarrow (1, 0) \rightarrow (0, 0)$
- **Tasks**
 - (a) Compute levels 1 and 2 of $\text{Sig}(\overleftarrow{X})$ with the A1 formula.
 - (b) Apply Chen's identity, $(A \otimes B)^{(2)} = A^{(2)} + A^{(1)}B^{(1)\top} + B^{(2)}$, to compute level 2 of $X * \overleftarrow{X}$.
 - (c) State what the result implies for the uniqueness of the signature.
- **Report:** the two signatures and level 2 of the concatenation; **Interpretation:** why do we add a time channel in practice?
- **Notebook:** section A3

A3 [Solved]: Solution

- (a) Reversed increments $(0, -2), (-1, 0)$: level 1 is $(-1, -2) = -S^{(1)}$; level 2: squares $\begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$ plus the pair $\begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}$, so $\overleftarrow{S}^{(2)} = \begin{bmatrix} 1/2 & 0 \\ 2 & 2 \end{bmatrix}$
- (b) Chen: $\begin{bmatrix} 1/2 & 2 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ -2 & -4 \end{bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$; level 1 is $S^{(1)} - S^{(1)} = 0$ as well
- (c) There and back is “tree-like”: its signature is that of the constant path; the signature identifies a path only up to such pieces [Hambly & Lyons \(2010\)](#)
- Interpretation: a strictly increasing time channel cannot go back, so tree-like pieces are impossible and the signature determines the path up to its starting point: we time-augment and supply the level separately

A4 [Proposed]: Lévy Area of a Loop — Task

Solution discussed in class

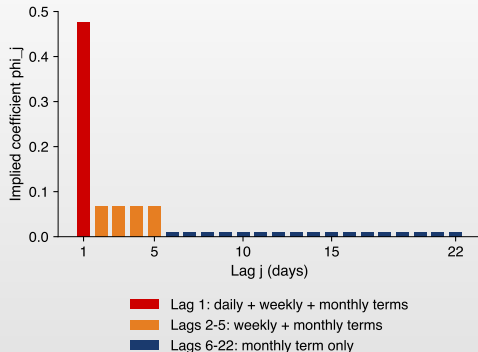
- ▣ **Question:** what does level 2 of the signature see in a closed loop that level 1 misses?
- ▣ **Model:** A3
- ▣ **Inputs:** The unit square $(0, 0) \rightarrow (1, 0) \rightarrow (1, 1) \rightarrow (0, 1) \rightarrow (0, 0)$, four linear segments
- ▣ **Tasks**
 - (a) Compute level 1 and the Lévy area A^{12} for the square run anticlockwise.
 - (b) Repeat for the square run clockwise, and for the square translated by $(5, 5)$.
 - (c) Explain why level 1 cannot distinguish the loop from a constant path while level 2 can.
- ▣ **Report:** three values of $S^{(1)}$ and A^{12} ; **Interpretation:** which invariance does the translation illustrate?
- ▣ **Notebook:** section A4

A5 [Solved]: HAR as a Restricted AR(22) — Task

- ▣ **Question:** which 22 lag coefficients does a HAR model for $\ln RV$ imply for JPM?
- ▣ **Inputs**
 - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only
 - ▶ HAR with *averages of logs*: $\ell_t = \ln RV_t$, $\ell_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 \ell_{t-j}$, $\ell_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} \ell_{t-j}$
 - ▶ The lecture's log-HAR uses logs of averages, $\ln RV_t^{(w)} \neq \ell_t^{(w)}$ (Jensen), and has no exact AR form
 - ▶ Full sample (2764 days), OLS, Newey–West (5 lags) s.e.: $\hat{\beta}_d = 0.408$ (0.027), $\hat{\beta}_w = 0.299$ (0.046), $\hat{\beta}_m = 0.196$ (0.040)
- ▣ **Tasks**
 - (a) Substitute $\ell_t^{(w)}$ and $\ell_t^{(m)}$ into the HAR equation and collect the coefficient ϕ_j of each lag ℓ_{t-j+1} , $j = 1, \dots, 22$.
 - (b) Compute $\phi_1, \phi_2 = \dots = \phi_5$ and $\phi_6 = \dots = \phi_{22}$ from the JPM estimates and plot them.
 - (c) Compute the sum $\sum_{j=1}^{22} \phi_j$.
- ▣ **Report:** the three distinct values, the sum, the chart; **Interpretation:** what does the sum measure, and is the process stationary?
- ▣ Notebook: section A5

A5 [Solved]: Solution

- (a) $\beta_w \ell_t^{(w)} = \sum_{j=1}^5 (\beta_w/5) \ell_{t-j+1}$ and $\beta_m \ell_t^{(m)} = \sum_{j=1}^{22} (\beta_m/22) \ell_{t-j+1}$: lag 1 collects all three terms, lags 2–5 the weekly and monthly ones, lags 6–22 only the monthly one
- (b) $\phi_1 = 0.408 + 0.299/5 + 0.196/22 = 0.476$; $\phi_{2..5} = 0.0688$; $\phi_{6..22} = 0.0089$
- (c) $\sum_j \phi_j = \beta_d + \beta_w + \beta_m = 0.903$
- Interpretation: the sum is the persistence of log RV; all $\phi_j \geq 0$ and the sum is below 1, so the process is stationary, with a slow, long-memory-like decay



A6 [Proposed]: Testing the HAR Restrictions — Task

Solution discussed in class

- ▣ **Question:** how do we test the restrictions that HAR imposes on an unrestricted AR(22)?
- ▣ **Model:** A5
- ▣ **Inputs**
 - ▶ The unrestricted AR(22) for $\ell_t = \ln RV_t$ has an intercept and slopes $\phi_1, \dots, \phi_{22}$; HAR with averages of logs (A5) is its restricted version
 - ▶ ϕ is the linear projection of ℓ_{t+1} on its 22 lags; weakly dependent data with finite fourth moments; errors may be heteroskedastic and serially correlated
- ▣ **Tasks**
 - (a) Count the linear restrictions and write them as $R\phi = 0$.
 - (b) Write the Wald statistic with an HAC covariance and its distribution under H_0 .
 - (c) State the 5% critical value.
- ▣ **Interpretation:** if the test rejects in a long sample, should we drop HAR for forecasting?
- ▣ **Notebook:** section A6

A8 [Proposed]: QLIKE With Its Arguments Swapped — Task

Solution discussed in class

- ▣ **Question:** which forecasts does a QLIKE with swapped arguments reward?
- ▣ **Model:** A7
- ▣ **Inputs:** A program computes $L(RV, F) = F/RV - \ln(F/RV) - 1$, QLIKE with the arguments swapped
- ▣ **Tasks**
 - (a) Find the forecast $F^\dagger$ that minimises $\mathbb{E}_t[L(RV, F)]$.
 - (b) Compare $F^\dagger$ with $\mathbb{E}_t[RV]$ using Jensen's inequality.
 - (c) Say which models the swapped loss favours.
- ▣ **Interpretation:** why can this bug pass elementary checks?
- ▣ **Notebook:** section A8

A10 [Proposed]: Limits of the Temperature — Task

Solution discussed in class

- ▣ **Question:** what happens to the kernel weights at the two extreme temperatures, and why scale by the median distance?
- ▣ **Model:** A9
- ▣ **Inputs:** n past windows, distances $\delta_\tau > 0$, temperature $\gamma = c/\text{median}(\delta)$ as in the lecture
- ▣ **Tasks**
 - (a) Find the limits of w and n_{eff} when $c \rightarrow 0$.
 - (b) Find the limits when $c \rightarrow \infty$ (assume a unique nearest window).
 - (c) Show that multiplying all distances by $k > 0$ leaves the weights unchanged.
- ▣ **Interpretation:** why is the median scaling needed to compare c across 50 assets?
- ▣ Notebook: section A10

B2 [Proposed]: The Lévy Area for E-mini S&P 500 Futures — Task

Solution discussed in class

- ▣ **Question:** does the Lévy area predict next month's volatility of E-mini S&P 500 futures (ES)?
- ▣ **Model:** B1
- ▣ **Inputs:** VOLARE, futures file, 28 Sep 2009 – 30 Jan 2026, ES; $rv5$, open and close prices; target, regressors and 22 Newey–West lags as in B1
- ▣ **Tasks**
 - (a) Repeat steps (a)–(d) of B1 for ES.
 - (b) Test the coefficient of A_t at 1% and 5%, two-sided.
 - (c) Compare the R^2 gain with the one for JPM.
- ▣ **Interpretation:** what does a positive coefficient say about windows with a negative area, and can the sign alone label them leverage episodes?
- ▣ **Notebook:** section B2

B4 [Proposed]: HAR, SHAR and HARQ for Crude Oil — Task

Solution discussed in class

- ▣ **Question:** do the SHAR asymmetry and the HARQ effect hold for crude oil futures (CL)?
- ▣ **Model:** B3
- ▣ **Inputs:** VOLARE, futures file, 28 Sep 2009 – 30 Jan 2026, CL; rv5, rsp5, rsn5, rq5; models, tests and Newey–West (5 lags) as in B3
- ▣ **Tasks**
 - (a) Estimate SHAR and test $\beta^- > \beta^+$ as in B3(b).
 - (b) Estimate HARQ and test $\beta_Q < 0$ as in B3(c); plot the coefficients with 95% HAC intervals.
 - (c) Compare both results with JPM.
- ▣ **Interpretation:** why might the asymmetry be stronger for a commodity than for a bank stock?
- ▣ **Notebook:** section B4

B6 [Proposed]: Sig-LK vs log-HAR for ES, One Week Ahead — Task

Solution discussed in class

- ▣ **Question:** does Sig-LK beat log-HAR for ES when the target is the average RV of the next five days?
- ▣ **Model:** B5
- ▣ **Inputs:** VOLARE, futures file, 28 Sep 2009 – 30 Jan 2026, ES; design of B5 with $h = 5$ and $c = 1.0$ (futures validation block)
- ▣ **Tasks**
 - (a) Produce weekly forecasts of Sig-LK and log-HAR with the design of B5.
 - (b) Compute the QLIKE ratio and its stationary-bootstrap 95% CI (mean block length 10).
 - (c) Run the DM test (bandwidth at least $h = 5$) and the conditional GW test; plot the cumulative loss difference.
- ▣ **Interpretation:** why is the CI so asymmetric, and what does it imply for the power of DM?
- ▣ **Notebook:** section B6

B8 [Proposed]: Multiple Testing at One Month Against HAR — Task

Solution discussed in class

- ▣ **Question:** on how many assets does Sig-LK beat HAR one month ahead after correcting for 50 tests?
- ▣ **Model:** B7
- ▣ **Inputs:** Lecture 20 results: 50 one-sided DM p -values for “Sig-LK better than HAR” at $h = 22$ (QLIKE)
- ▣ **Tasks**
 - (a) Count the rejections at 5% without correction.
 - (b) Apply Holm and BH as in B7, count the rejections and name the asset with the smallest p -value.
 - (c) Compare with the pooled mean QLIKE ratio of Sig-LK relative to HAR at $h = 22$ (1.334).
- ▣ **Interpretation:** how can Sig-LK beat HAR on several assets and still lose on average?
- ▣ **Notebook:** section B8

C1 [Proposed]: A Level-Aware Signature Kernel — Task

Solution discussed in class

- ▣ **Question:** does a kernel that also sees the *level* of volatility make the weights useful in a crash?
- ▣ **Inputs**
 - ▶ The kernel of the lecture is translation-invariant, so it cannot recognise a high-volatility regime (Lecture 20, weights on 16 March 2020); data: VOLARE, the 40 stocks, 2015–2026
 - ▶ Idea: prepend a **basepoint** [Morrill et al. \(2020\)](#): the path starts at the training mean of $\ln RV$ and jumps to the first value of the window; by Chen, $\text{Sig} = \exp_{\otimes}(v) \otimes \text{Sig}(X)$ with v the standardised level jump
- ▣ **Tasks**
 - (a) Write a one-page pre-registration: 40 stocks, $h = 1$, the lecture design, $c = 0.5$, QLIKE, DM with Holm, the COVID-19 subperiod reported separately.
 - (b) Implement the basepoint kernel (use the Chen update of the notebook) with the same features as Sig-LK, and forecast.
 - (c) Compare with Sig-LK and log-HAR: mean QLIKE ratios (all days and COVID-19), wins over log-HAR raw and after Holm.
- ▣ **Interpretation:** if seeing the level does not help, where would you look next (features, target, horizon)?
- ▣ Notebook: section C1

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Applied Time Series Analysis with Python

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20; data: volare.unime.it

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